ZEBA AFREEN

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SUMMARY

Computer Science graduate student who works from relational databases through model evaluation and treats every result as something to verify before trusting it. Brings habits of held-out testing, sensitivity analysis, and pre-release validation, and turns requirements from non-technical stakeholders into clear, explainable results.

EXPERIENCE

IT Intern | *Gates Engineering & Services, Dubai, UAE* **March 2024 – June 2024**

- Created and modified 15+ Oracle SQL views, queries, and stored procedures for ERP reporting, turning Customer Service and Sales requirements into reports.
- Tested and validated ERP workflow changes with the IT team, third-party developers, and 2 interns, running test scenarios to catch inconsistencies before production deployment.
- Resolved 6–10 end-user issues per week via the ticketing system across hardware, software, network, and ERP systems.

Web Development Intern | *iCodeJr, Dubai, UAE* **July 2023 – September 2023**

- Rebuilt the company's early-stage WordPress website, delivering 5+ core pages and age-specific program and syllabus pages with WordPress plugins and HTML.

PROJECTS

Spotify Audio Feature Analysis & Generative Modeling | *Python, PCA, t-SNE, UMAP, GMM, VAE* **May 2026**

- Analyzed ~114,000 tracks across 114 genres using Pearson/Spearman correlation and mutual information to identify the most informative audio features.
- Applied PCA, t-SNE, UMAP, and Gaussian Mixture Modeling to find interpretable clusters across 10 standardized audio features.
- Trained a 3-dimensional Variational Autoencoder for 50 epochs to generate synthetic song features while preserving the source data's major correlation patterns.

Cloud vs. Owned Infrastructure Cost & Scaling Analysis | *AWS, Cost Modeling* **May 2026**

- Led the Netflix case study in a five-year cloud-vs-owned analysis, modeling a workload of ~2,400 baseline instances across stable, spiky, and high-growth scenarios.
- Developed the server-count methodology and audited assumptions across Shopify- and Netflix-scale models covering compute, storage, data transfer, hardware, and power costs.

Brain Tumor Segmentation Using Multi-Modal MRI | *Python, U-Net, Deep Learning, Computer Vision* **December 2025**

- Built a U-Net multi-class segmentation pipeline on BraTS 2020, processing 369 patient MRI volumes and 57,000+ 2D slices across 4 MRI modalities.
- Trained with combined Dice and categorical cross-entropy loss, data augmentation, and early stopping, reaching ~0.88 validation Dice with stable convergence.

Online Auction System | *Python, Flask, MySQL, HTML/CSS* **May 2025**

- Developed a full-stack relational auction platform with listings, search, bidding, bid histories, customer-service workflows, and administrative sales reporting.
- Implemented an automatic bidding engine with user-defined upper limits and alerts, with role-based access for 3 user types: end users, customer representatives, and administrators.

EDUCATION

Rutgers University | *M.S. in Computer Science, New Brunswick, NJ* **Expected December 2026**

Manipal Academy of Higher Education | *Dubai, UAE* **July 2024**

Bachelor of Technology in Computer Science, Minor in Computational Intelligence | GPA: 9.20 / 10

LANGUAGES & TECHNOLOGIES

Languages & Databases: Python, Java, C++, JavaScript, SQL, MySQL, Oracle SQL, MATLAB

ML & Data: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, Matplotlib

Software & Cloud: Flask, React, HTML/CSS, AWS, Git, GitHub, Stored Procedures, Data Structures & Algorithms

CERTIFICATIONS & ACTIVITIES

- Certifications: Mathematics for Machine Learning: Linear Algebra (Coursera); IoT Fundamentals: Connecting Things (Cisco); Introduction to Cybersecurity (Cisco).
- Co-authored and presented an IoT healthcare paper at AEIT 2024; competed in IEEE's 24-hour IEEEXtreme 17.0 programming challenge.